

Malthe Sparring

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Research interests

Homotopy theory, (∞) -category theory, algebraic topology, factorisation algebras.

Education

University of Edinburgh

Sep 2022 - Aug 2027

PhD Geometry and Topology

(Expected)

- Working with Prof. Clark Barwick on a project on " n -fold monoidal ∞ -categories."
- (Sep - Oct 2025) Received grants from the London Mathematical Society and the Institute for Mathematics and its Applications for a two-month research visit to NTNU Trondheim.

University of Edinburgh

Sep 2017 - May 2022

MMath Mathematics

- Awarded a 2021 Summer Vacation Scholarship for a project on "Axiomatic Homology and the Borsuk-Ulam Theorem".

Invited talks

- **Pre-monoidal ∞ -operads** September 2026
Universitat Autònoma de Barcelona Topology Seminar
- **The pre-monoidal envelope of an ∞ -operad** (Poster; available [here](#).) June 2026
Young Topologist Meeting, Copenhagen
- **Twofold monoidal ∞ -categories** March 2026
Warwick University Junior Algebra Seminar
- **Twofold monoidal structures** May 2025
University of Aberdeen Postgraduate Research Seminar

Teaching

Mathematics Tutor

Jan 2023 - Current

University of Edinburgh

- (Autumn 2026) Introduction to Mathematics at University
- (Spring 2026) Honours Algebra, Honours Algebra - Skills (Mathematical computing).
- (Spring 2025) Calculus and its Applications.
- (Autumn 2024) Accelerated Proofs and Problem Solving, Axiomatic Set Theory, MathsBase (Drop-in study space).
- (Spring 2024) Honours Algebra - Skills (Mathematical computing), Proofs and Problem Solving.
- (Autumn 2023) Category Theory, Honours Analysis - Skills (Presentation skills), MathsBase (Drop-in study space).
- (Spring 2023) Algebraic Topology, Fundamentals of Pure Mathematics.

Fall 2023 and 2025

Volunteering

- Mentored two projects for [Twoples](#), an online directed reading program for mathematics undergraduates. The first project was on algebraic topology, the second project is on homological algebra.

Professional Experience

Online Assessment Intern

July - August 2025

University of Edinburgh

- Took part in a project to turn the HELM mathematics workbooks into STACK quizzes. This included

managing interns and performing quality assurance of their work. STACK is an online assessment system for mathematics and STEM.

Online Assessment Intern

Aug - Sep 2022

University of Glasgow

- Responsible for converting a series of diagnostics tests for undergraduate students into the STACK system.

Online Assessment Intern

May - June 2022

Heriot-Watt University

- Took part in a project to introduce STACK assessment in first year mathematics courses. This included designing new questions, and running a workshop for colleagues.

Freelance Mathematics Tutor

Jan - May 2022

- Gained experience tutoring students age 16-18 of the International Baccalaureate programme, in both an individual and group setting.

ASID Intern

June - Aug 2020

University of Edinburgh

- Took part in a project to turn the HELM mathematics workbooks into STACK quizzes. This included conducting quality-assurance on questions written by colleagues.
- Gained experience in teaching STACK from organizing and speaking at eight STACK workshops.
- Learned HTML and CSS through designing the official website stack-assessment.org.

Online Assessment Community Engagement Intern

June - Aug 2019

University of Edinburgh

- Responsible for improving the learning experience of the online mathematics assessment system STACK.
- Collated a [a collection of cases studies](#) detailing innovative use of STACK.
- Conducted a survey of STACK users world-wide.

Technologies

Experienced user of Python, Git, Lean and LaTeX.

Experienced in online mathematics assessment via STACK